Tutorial No.: 9

Engineering Mechanics (ME-101)

Date: 26/03/2024 and Time: 7.55 to 8.55 AM

Question 1: Each of the three balls in Figure 1, has mass m and is welded to the rigid equiangular frame of negligible mass. The assembly rests on a smooth horizontal surface. If a force F is suddenly applied to one bar shown, determine (a) the acceleration of the point O and (b) the angular acceleration $\ddot{\theta}$ of the frame.

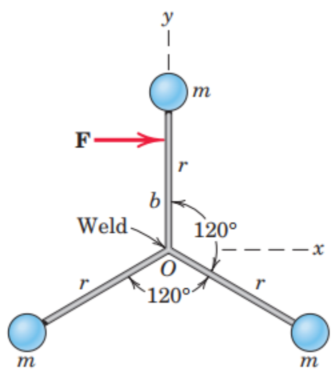

Figure 1

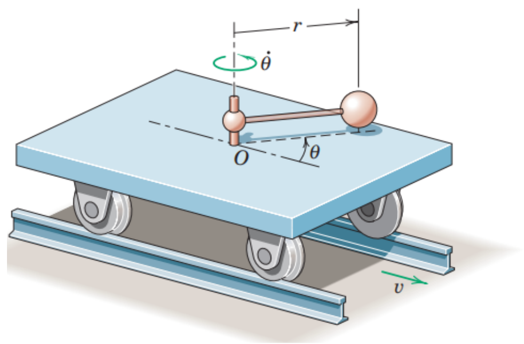

Figure 2

Question 2: The small car shown in Figure 2, has a mass of 20 kg, rolls freely on the horizontal track and carries the 5-kg sphere mounted on the light rotating rod with $r = 0.4$ m. A geared motor drive maintains a constant angular speed $\theta = 4$ rad/s of the rod. If the car has a velocity $v = 0.6$ m/s when $\theta = 0°$, calculate v when $\theta = 60°$. Neglect the mass of the wheels and any friction.

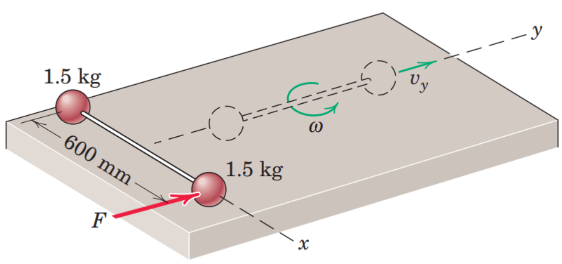

Figure 3

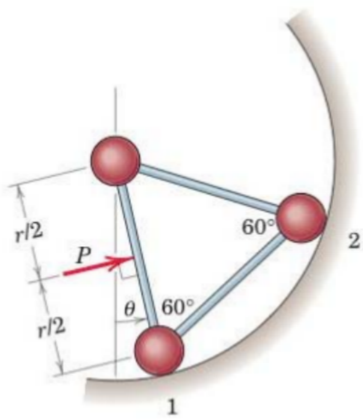

Figure 4

Question 3: The two spheres, depicted in Figure 3, are rigidly connected to the rod of negligible mass and are initially at rest on the smooth horizontal surface. A force F is suddenly applied to one sphere in the y-direction and imparts an impulse of 10 N-s during a negligibly short period of time. As the spheres pass the dashed position, calculate the velocity of each one.

Question 4: The three small spheres, each of mass m, are secured to the light rods to form a rigid unit supported in the vertical plane by the smooth circular surface (refer Figure 4). The force of constant magnitude P is applied perpendicular to one rod at its midpoint. If the unit starts from rest at $\theta = 0$, determine (a) the minimum force P_{min} which will bring the unit to rest at $\theta = 60^0$, and (b) the common velocity v of spheres 1 and 2, when $\theta = 60^0$, if $P = 2P_{min}$.

Question 5: As shown in Figure 5, a high speed jet of air issues from the nozzle A with a velocity of v_A and mass flow rate of 0.36 kg/s. The air impinges on a vane causing it to rotate to the position shown. The vane has a mass of 6 kg. Knowing that the magnitude of the air velocity is equal at A and B, determine, (a) the magnitude of the velocity at A, and (b) the components of the reactions at O.

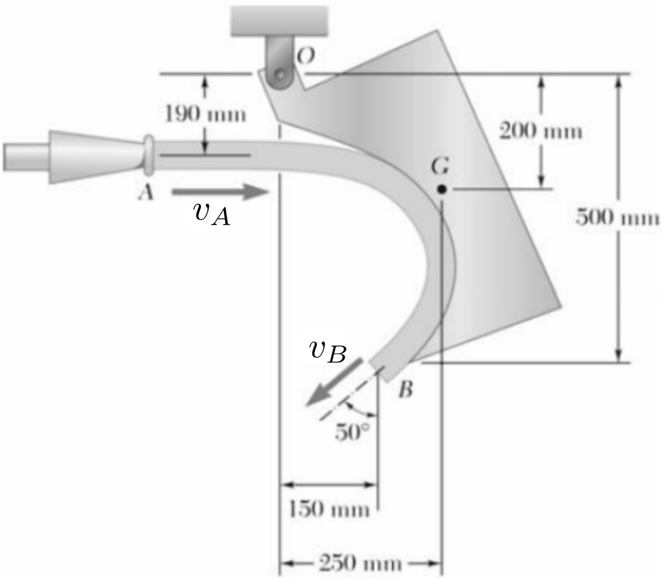

Figure 5

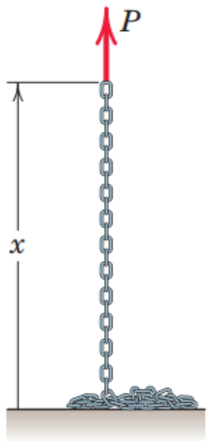

Figure 6

Question 6: The end of a chain of length L and mass ρ per unit length which is piled on a platform is lifted vertically with a constant velocity v by a variable force P (refer Figure 6). Find P as a function of the height x of the end above the platform. Also find the energy lost during the lifting of the chain.